Sahaj Bhadja

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Summary

Software developer with 2+ years designing, maintaining, and troubleshooting **secure research and business applications**, web services, integrations, and SQL Server systems; experienced in Python, JavaScript, PHP, testing, automation, technical consulting, and documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Designed and maintained **secure Python applications and relational schemas** using FastAPI, Azure SQL, PostgreSQL, and SQL Server, supporting 500+ university users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Ensured **daily operational integrity for production services** processing 5,000+ monthly jobs, troubleshooting complex data and integration errors with validation, reconciliation, explicit states, documentation, and CI/CD-backed fixes achieving 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Analyzed user requirements and developed **React, JavaScript, and Python web applications**, building Flask services for uploads, processing, history, and reporting that supported 150+ users and 10,000+ daily requests.
- Tested and debugged **moderate-to-complex application logic** across LLM, vision, OCR, and file-processing integrations, implementing secure validation, fallback handling, persistent results, and fault-tolerant execution while reducing turnaround time 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Developed **Python simulation and evaluation pipelines** for university research, modeling autonomous-system trajectories from map geometry and temporal signals while translating research objectives into reproducible technical workflows and benchmarks.
- Analyzed model behavior and documented **metrics, system constraints, and failure patterns**, collaborating with researchers to test hypotheses, evaluate software outputs, and refine experimental configurations for reliable comparative analysis.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed and configured **Python and Java application integrations** using REST APIs, SQL, caching, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Collaborated with Microsoft and GitHub stakeholders to define scope, investigate defects, document solutions, and deliver version-controlled enhancements through Jenkins, GitHub Actions, and 20+ automated unit and integration tests.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Designed and improved **user-focused React, JavaScript, and Node.js applications** across eight services for authentication, records, billing, permissions, and notifications, integrating 2M+ records through REST APIs, WebSockets, and Azure SQL.
- Maintained production integrity through **automated testing and troubleshooting** with Jest, Selenium, Sentry, Azure Monitor, and Kibana, deploying Docker releases via Jenkins CI/CD while reducing launch time 50%.

Skills

Languages: Python, JavaScript, PHP, Java, TypeScript, SQL, Bash, HTML, CSS
Web & Integration: Spring Boot, FastAPI, Flask, Django, Node.js, React.js, REST APIs, web services, microservices
Databases: Microsoft SQL Server, Azure SQL, PostgreSQL, MySQL, Oracle, schemas, data manipulation, query optimization
Automation & AI: Backend data pipelines, scripting, CLI, Linux, ChatGPT, Claude, LLM integrations, AI-assisted workflows
Engineering: Git, GitHub, Jenkins, CI/CD, Docker, testing, debugging, systems analysis, technical documentation, Agile/Scrum

Projects

Chat2SQL Code | *Java, TypeScript, Oracle, Docker, Maven*
- Enhanced an **AI-assisted database application** across Java and TypeScript components, supporting Oracle and 10+ SQL/NoSQL systems with schema exploration, query generation, reporting, Docker, Maven, and cross-platform troubleshooting workflows.

HMIS Live | *MERN, Microservices, OOP*
- Developed a **secure healthcare administration application** with role-based workflows, REST integrations, TypeScript dashboards, notifications, and Sentry monitoring, halving record retrieval time while reducing administrative workload 30% through streamlined workflows.

Big-Brain Live | *Python, React, PostgreSQL, REST APIs*
- Built an **AI-assisted document and reporting application** processing 50,000+ records with Python pipelines, PostgreSQL metadata, REST APIs, authentication, incremental indexing, and system maintenance workflows, reducing retrieval latency 40%.